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# EXTRUSION CHEAT SHEET EXTENDED INSTRUCTIONS

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To be used in combination with the cheat sheet

## Step A : before beginning

This workflow uses custom g-code that also needs to be adapted during the process. Always be careful when printing with custom code, the machine might do things you did not intend.

The code and sheet should be adapted to the print bed size. Print beds bigger or equally big as the ender 3V2 printbed could still use the same size. For the sheet to be as effective as possible however, the cheat sheet should be scaled to as big as possible.

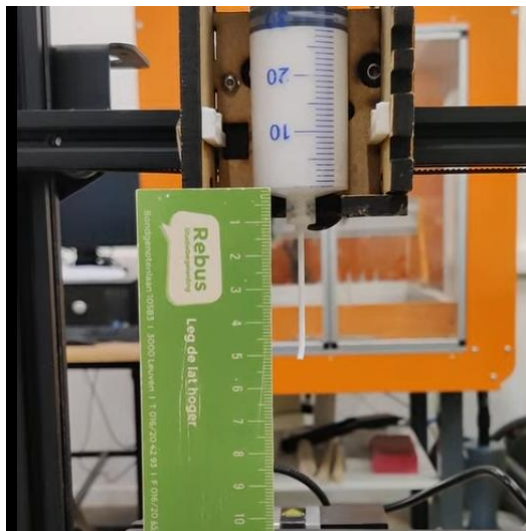
Depending on the size of the nozzle you will also need to scale the sheet. This scale works for nozzles ranging from 0.4 mm to 5 mm.

## Step B : using the cheat sheet

### Step 0 – E steps

Cut off the information piece of the E steps and tape / glue / clamp the sheet on the print bed.

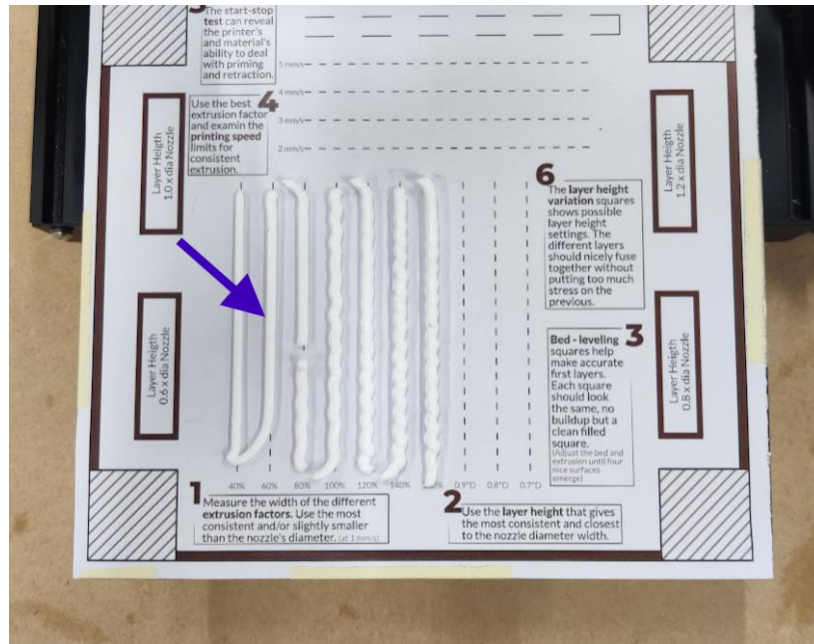
- ⇒ Calculate the theoretical g-code and input this in the machine settings (under advanced settings)
- ⇒ Move the extrusion nozzle upwards in Z
- ⇒ Command an amount of extrusion movement (not too much or gravity will thin the extrudate)
- ⇒ Measure the resulting extrudate
- ⇒ Fine-tune the E-steps value in your machine settings



## Step 1 – Extrusion factors

The Z height for this test should be within a free depositing regime, the Z height should be about 1.5 times the nozzle diameter.

- ⇒ Adapt Z height in gcode, save and upload to printer
- ⇒ Observe and measure the resulting lines, the most consistent, straight and smoothest line is the value to select

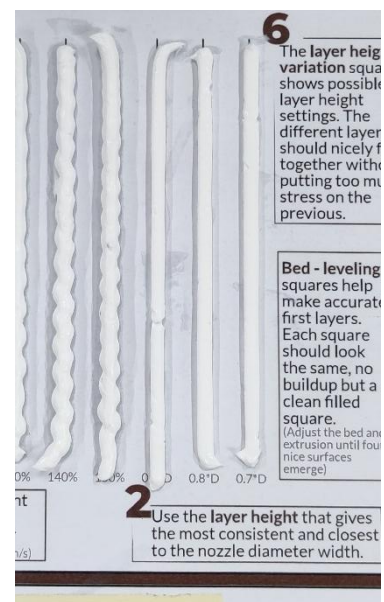


In this example 60% extrusion factor gave the most constant width (first inspect visually then measure with callipers)

## Step 2 – Layer height

- ⇒ Adapt the Z height in the gcode with your nozzle diameter dimensions
- ⇒ Adapt the extrusion speed with the factor found in the previous step, save and upload to printer
- ⇒ Measure and choose the most consistent, filled in line.  
This measured width will be the line width of your extrudates later

In this example 0.7\*D is the best factor resulting in a line width of 4 mm.



## Step 3 – Bed-levelling

This is the step where the most adaption is needed in the gcode, but it is also a step that might not be necessary if your printer has auto bed-levelling.

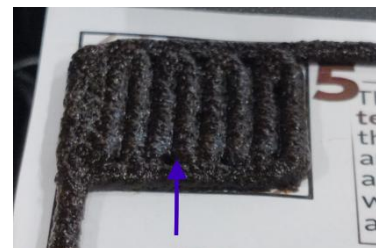
The adaption that needs to happen to the gcode is changing the spacing between the infill lines to be equal to the line width found in the previous test. This is no longer just changing one value, all coordinates of the infill lines need to be adapted for this if your line width differs from 4 mm. A helpful tool to fulfil this task are large language models, these are quite good at the repetitive task of changing this gcode. Used together in combination with an online NC / gcode viewer to check the resulting gcode the daunting task of changing this gcode gets a lot easier.

- ⇒ Change the gcode with the found line width and layer height
- ⇒ Observe the corner rectangles for over and under extrusion in the infill patterns
- ⇒ Corners that get over extruded need to be lowered, corners with under extrusion need to be lifted
- ⇒ The long leading lines show how your printer deals with longer continuous extrusion



Slight over extrusion seen by extra material

Under extrusion seen by valley and ridges



## Step 4 – Printing speed

- ⇒ Adapt the gcode with the found extrusion speed and layer height
- ⇒ Choose the fastest printing speed that delivers a consistent extrudate

The lines may all be consistent but also listen and look at your printer while doing this step. If the machine seems to be struggling, it might be wiser to choose a lower value even though the lines seem okay.

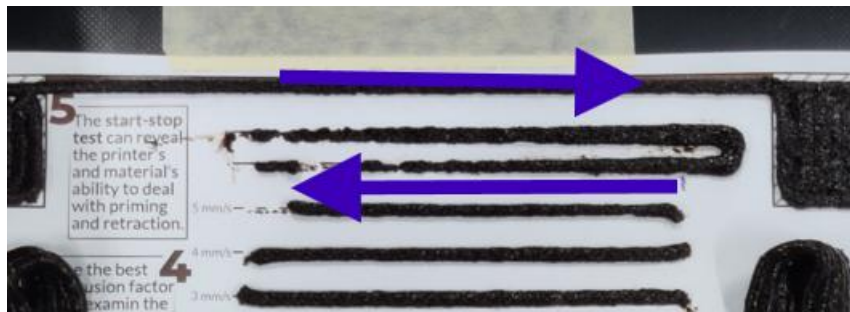
It is possible to later redo this test with even higher speeds if all lines were printed with ease.

## Step 5 – Start stop test

The start stop test shows the ability to printer shorter segments. It is an indication of how long your setup takes to prime the nozzle and to generate a full width line. It also shows how much the material and set up experiences coasting (material keeps coming out of the nozzle even though the extruder motor has stopped). This happens due to residual stress being build up. If your setup and material allow retraction then this setting, combined with priming setting and coasting settings can be used to tweak this phenomenon.

- ⇒ Adapt the Z layer height, E extrusion speed and F printing speed in the gcode, save and upload to your printer
- ⇒ Observe how long it takes for the printer to reach a line with full width
- ⇒ Observe how long materials keeps coming out of the nozzle

Priming distance



Coasting distance

## Step 6 – Layer height towers

- ⇒ Again adapt the gcode with the found layer height, extrusion speed and printing speed. Be careful, the code needs to be adapted on all layers of all four the rectangles
- ⇒ Save the code and upload it to your printer
- ⇒ Observe the rectangles and pick the stable, good bonded layers

A smaller layer height  $0.6 \times \text{width}$  might squish the printed part too much but will give good bonding. A bigger layer height  $1.2 \times$  might give worse layer bonding but will enable quicker printing of larger parts. The five layer high structures also already indicate how the material will behave when stacked.

## Step C : what now

Enjoy printing with your new material!

### Follow up tests

From this point onwards further steps are heavily dependent of the material being used, the geometry of the print, the setup being used, the wanted results, ...

This is a list of what I believe to be the most useful following tests but definitely not for each case:

- Overhang / bridging test – unsupported spans, probes yield stress and buildability
- Interlayer time-interval / drying test – varies pause time between layers to find the drying/curing window needed for good adhesion
- Wall-thickness / single-perimeter test – extrusion consistency through corners and direction changes
- Infill / multi-strand bonding test – consistency and bonding between adjacent, not just stacked, strands
- Shrinkage / dimensional-stability test – printed vs. dried/cured dimensions.
- Overhang-angle / support-need test – maximum printable angle before supports are required
- Retraction/stringing test – material behaviour during travel moves between separate features
- Fine-detail / resolution test – smallest reliably printable feature size (thin walls, small text, sharp corners)
- Benchmark object print (e.g. a 3DBenchy-style or custom test geometry) – combines several of the above into one representative print

### Slicer settings

I recommend going towards the following site: <https://wikifactory.com/@jonathankeep>

Jonathan Keep is an expert in clay printing and has a detailed guide of how to use and setup Cura slicer for clay printing. This setup can be extracted and generalised towards all custom extrusion printing